Student Hub Portal

Practical File

Practicals 1 to 5 - HTML, CSS and JavaScript

Practical-wise write-up with questions, answers, analysis, interpretation, and source code.

Student Hub Portal Source Code

Practical No. 1 - Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Question: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

Key Questions

Q: What is URL and parts of URL?

Answer: A URL is the address used to open a resource on the internet. Its main parts are the protocol such as https, domain name, path, query string, and fragment.

Q: How does HTML file processed in web browser?

Answer: The browser reads the HTML file, builds the DOM tree, loads linked CSS and JavaScript files, and then displays the final page on the screen.

Q: How will page navigation flow managed among all HTML pages?

Answer: Navigation is managed with anchor tags and relative links such as index.html, about.html, events.html, and register.html. A common navigation bar keeps the flow consistent.

Q: How will GitHub commits be maintained after each practical?

Answer: After completing and testing each practical, the changed files should be added, committed with a clear message, and pushed to the GitHub repository.

Analysis

A: The Student Hub project is divided into clear pages for home, information, events, support, authentication, student details, and administration. Separate HTML, CSS, and JavaScript files keep the work organized.

Interpretation

I: A sitemap and wireframe give the project a clear direction before coding starts. This reduces confusion about page connections and makes later GitHub commits easier to understand.

Project File Structure

```
index.html
about.html
events.html
faq.html
contact.html
login.html
register.html
dashboard.html
profile.html
admin.html
style.css
script.js
```

Practical No. 2 - Semantic HTML5 Pages with Accessibility-Ready Structure

Question: Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

Key Questions

Q: Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Answer: Yes. The pages use header for branding, nav for links, main for page content, sections or cards for grouped information, and footer for common bottom links.

Q: Are form labels, headings, image alt text, and navigation links accessible?

Answer: The form fields have labels, the pages use headings, the logo has alt text, and navigation is provided through normal links. These features help keyboard and screen-reader users understand the page.

Q: Is the page structure consistent across all pages?

Answer: Yes. The pages follow a common pattern with a header, navigation menu, main content area, footer, and linked CSS and JavaScript files.

Analysis

A: The project has ten linked HTML pages with a common layout. Semantic elements make the structure easier for browsers, developers, and assistive technologies to understand.

Interpretation

I: The pages form a consistent static website foundation. A user can move from the home page to information, events, registration, login, and support pages without losing the common site structure.

HTML Code - index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Home</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- 1. Notification Banner -->
  <div id="notification-banner" class="notification-banner">
    <span>📢 <strong>Notice:</strong> Tech Fest 2026 registration is now open!</span>
    <button class="banner-close" onclick="closeBanner()">&times;</button>
  </div>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <!-- 2. Light / Dark Theme Switcher -->
      <button onclick="toggleTheme()">🎨 Theme</button>
```

```
    <button><a href="login.html">Login</a></button>
    <button><a href="register.html">Register</a></button>
  </div>
</header>

<!-- 3. Hamburger Menu & Navigation -->
<div class="navlinks">
  <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
  <nav id="nav-menu" align="center">
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="events.html">Events</a>
    <a href="faq.html">FAQ</a>
    <a href="contact.html">Contact</a>
  </nav>
</div>

<main>
  <div class="main">
    <h1>Student Hub Portal</h1>
    <p>Manage Students &amp;amp; College Events Easily</p>
    <br>
    <!-- 4. Modal Popup Trigger -->
    <button onclick="openModal('welcomeModal')">View Announcement</button>
  </div>

  <!-- 5. Content / Announcement Slider -->
  <div class="slider-container">
    <div class="slide">
      <h3>🎉 Welcome New Students!</h3>
      <p>Orientation program starts this Monday at 10:00 AM in the Auditorium.</p>
    </div>
    <div class="slide">
      <h3>⏳ Annual Sports Meet 2026</h3>
      <p>Register your sports teams before October 20th to participate.</p>
    </div>
    <div class="slide">
      <h3>💡 Hackathon Submissions Open</h3>
      <p>Submit your software project ideas on the dashboard before Friday.</p>
    </div>
    <div class="slider-controls">
      <button onclick="changeSlide(-1)">❮ Prev</button>
      <button onclick="changeSlide(1)">Next ❯</button>
    </div>
  </div>

  <br>

  <div style="display: flex; gap: 20px;">
    <div class="box">
      <h3 style="font-size:24px; text-decoration: underline;">Latest Notice</h3>
      <ul>
        <li class="t1">Semester Exam Schedule Released</li>
        <li class="t1">Holiday on Friday</li>
        <li class="t1">Library Books Return Date</li>
      </ul>
    </div>

    <div class="box">
      <h3 style="font-size:24px; text-decoration: underline;">Upcoming Events</h3>
      <ul>
        <li class="t1">Tech Fest - Oct 25</li>
        <li class="t1">AI Workshop - Nov 10</li>
        <li class="t1">Career Seminar - Dec 02</li>
      </ul>
    </div>
```

```
      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Quick Links</h3>
        <ul>
          <li class="t1links"><a href="dashboard.html">Student Portal</a></li>
          <li class="t1links"><a href="events.html">Events</a></li>
          <li class="t1links"><a href="faq.html">FAQs</a></li>
        </ul>
      </div>
    </div>
  </main>

  <!-- 6. Modal Popup Structure -->
  <div id="welcomeModal" class="modal">
    <div class="modal-box">
      <button class="modal-close" onclick="closeModal('welcomeModal')">&times;</button>
      <h2 style="color: #EC9F05; margin-bottom: 10px;">Important Announcement</h2>
      <p style="margin-bottom: 15px;">Welcome to the Student Hub Portal! Check the Events
section for upcoming workshops and register for campus activities.</p>
      <button onclick="closeModal('welcomeModal')">Got it!</button>
    </div>
  </div>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <!-- JavaScript File -->
  <script src="script.js"></script>
</body>

</html>
```

HTML Code - about.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - About</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
```

```
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 900px; margin: 0 auto; text-align: center;">
    <h1 style="font-size: 32px; margin-bottom: 20px;">About Student Hub</h1>

    <div style="display: flex; gap: 20px;">
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Mission</p>
        <p style="font-size: 18px;">To eliminate communication gaps across campus by
delivering real-time updates and effortless management tools for every student.</p>
      </div>
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Vision</p>
        <p style="font-size: 18px;">To build a modern, fully digital campus ecosystem where
information is accessible anywhere, anytime with zero operational friction.</p>
      </div>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - events.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Events</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
```

```
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 900px; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 20px;">College Events</h1>

    <!-- Search Bar -->
    <div class="table1" style="padding: 15px; text-align: center;">
      <label for="search" style="font-size: 16px; margin-right: 10px;"><strong>Search
Events:</strong></label>
      <input type="text" id="search" placeholder="Search events by name or category..."
style="max-width: 350px; margin: 8px auto; display: inline-block;">
      <button style="margin-left: 5px;">Search</button>
    </div>

    <br>

    <!-- Events List Container -->
    <div class="table1">
      <table width="100%" cellpadding="15" style="border-collapse: collapse;">

        <!-- Event 1 -->
        <tr>
          <td width="20%" align="center">
            <div style="width: 100px; height: 80px; background-color: #0B162C; border: 2px
solid #6B4EFD; border-radius: 12px; display: flex; align-items: center; justify-content:
center; color: #EC9F05; font-weight: bold;">
              SPORTS
            </div>
          </td>
          <td width="60%">
            <h3 style="font-size: 22px; color: #EC9F05; margin-bottom: 6px;">SPOURAL -
2026</h3>
            <p><strong>Date:</strong> October 25, 2026</p>
            <p><strong>Venue:</strong> College Main Ground</p>
            <p><strong>Category:</strong> Annual Sports Festival</p>
          </td>
          <td width="20%" align="center">
            <button onclick="openModal('eventModal1')">Register</button>
          </td>
        </tr>

        <tr>
          <td colspan="3">
            <hr style="border-color: #6B4EFD; margin: 10px 0;">
          </td>
        </tr>

        <!-- Event 2 -->
        <tr>
          <td width="20%" align="center">
            <div style="width: 100px; height: 80px; background-color: #0B162C; border: 2px
solid #6B4EFD; border-radius: 12px; display: flex; align-items: center; justify-content:
center; color: #EC9F05; font-weight: bold;">
              TECH
            </div>
          </td>
          <td width="60%">
            <h3 style="font-size: 22px; color: #EC9F05; margin-bottom: 6px;">Cyber Security
Workshop</h3>
            <p><strong>Date:</strong> November 10, 2026</p>
```

```
            <p><strong>Venue:</strong> Auditorium B</p>
            <p><strong>Category:</strong> Hands-on Technical Workshop</p>
          </td>
          <td width="20%" align="center">
            <button onclick="openModal('eventModal2')">Register</button>
          </td>
        </tr>

      </table>
    </div>
  </main>

  <!-- Event 1 Modal -->
  <div id="eventModal1" class="modal">
    <div class="modal-box">
      <button class="modal-close" onclick="closeModal('eventModal1')">&times;</button>
      <h2 style="color: #EC9F05; margin-bottom: 10px;">Register for SPOURAL-2026</h2>
      <p style="margin-bottom: 15px;">Are you ready to participate in the Annual Sports
Festival? Click confirm to book your slot.</p>
      <button onclick="alert('Successfully registered for SPOURAL-2026!');
closeModal('eventModal1')">Confirm Registration</button>
    </div>
  </div>

  <!-- Event 2 Modal -->
  <div id="eventModal2" class="modal">
    <div class="modal-box">
      <button class="modal-close" onclick="closeModal('eventModal2')">&times;</button>
      <h2 style="color: #EC9F05; margin-bottom: 10px;">Register for Cyber Security</h2>
      <p style="margin-bottom: 15px;">Join the hands-on ethical hacking & security workshop.
Limited seats available.</p>
      <button onclick="alert('Successfully registered for Cyber Security Workshop!');
closeModal('eventModal2')">Confirm Registration</button>
    </div>
  </div>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - faq.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - FAQ</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
```

```
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 800px; margin: 0 auto; padding: 15px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 20px;">Frequently Asked
Questions</h1>

    <!-- Collapsible FAQ Accordion -->
    <div class="table1">
      <div class="faq-item">
        <button class="faq-btn" onclick="toggleFaq(this)">
          <span>Q. How do I register as a student?</span>
          <span class="faq-icon">+</span>
        </button>
        <div class="faq-answer">
          <p>Click on the 'Register' button in the top menu, fill in your student details
(name, email, course, semester), and click submit.</p>
        </div>
      </div>

      <div class="faq-item">
        <button class="faq-btn" onclick="toggleFaq(this)">
          <span>Q. How do I login to my dashboard?</span>
          <span class="faq-icon">+</span>
        </button>
        <div class="faq-answer">
          <p>Go to the Login page, enter your registered email address and password, then
click the 'Login' button.</p>
        </div>
      </div>

      <div class="faq-item">
        <button class="faq-btn" onclick="toggleFaq(this)">
          <span>Q. How can I join college events?</span>
          <span class="faq-icon">+</span>
        </button>
        <div class="faq-answer">
          <p>Navigate to the Events page, browse available workshops or competitions, and
click 'Register' for your selected event.</p>
        </div>
      </div>

      <div class="faq-item">
        <button class="faq-btn" onclick="toggleFaq(this)">
          <span>Q. How do I contact the college admin?</span>
          <span class="faq-icon">+</span>
        </button>
        <div class="faq-answer">
          <p>You can use the Contact page form or send an email through the contact section to
submit your questions directly to the admin.</p>
```

```
      </div>
    </div>
  </div>
</main>

<div class="footer">
  <footer align="center">
    <p>
      <a href="about.html">About</a>
      <a href="contact.html">Contact</a>
      Copyright &copy; Student Hub Portal
    </p>
  </footer>
</div>

<script src="script.js"></script>
</body>

</html>
```

HTML Code - contact.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Contact &amp;amp; Feedback</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 900px; margin: 0 auto; padding: 20px;">

    <!-- Contact Info Card -->
    <div class="table1" style="padding: 25px;">
      <h2 style="font-size: 28px; margin-top: 0; color: #EC9F05;">Contact Information</h2>
      <hr style="border-color: #6B4EFD; margin: 15px 0;">
      <p style="font-size: 18px;"><strong>Email :</strong> support@studenthub.com</p>
      <p style="font-size: 18px;"><strong>Phone :</strong> +1 234 567 8900</p>
      <p style="font-size: 18px;"><strong>Address :</strong> Campus Center, Student Block
        4</p>
```

```
    </div>

    <br>

    <!-- Feedback Form Card -->
    <div class="table1" id="feedback" style="padding: 25px;">
      <h2 style="font-size: 28px; margin-top: 0; color: #EC9F05;">Feedback</h2>
      <form action="#">
        <p>
          <label for="fb-name"><strong>Name</strong></label><br>
          <input type="text" id="fb-name" required style="width: 100%; margin-top: 6px;">
        </p>
        <p>
          <label for="fb-email"><strong>Email</strong></label><br>
          <input type="email" id="fb-email" required style="width: 100%; margin-top: 6px;">
        </p>
        <p>
          <label for="fb-message"><strong>Message</strong></label><br>
          <textarea id="fb-message" rows="4" required style="width: 100%; margin-top:
6px;"></textarea>
        </p>
        <p align="center" style="margin-top: 20px;">
          <button type="submit" style="padding: 10px 30px; font-size: 18px;">Send</button>
        </p>
      </form>
    </div>
  </main>
  <br>
  <div class="footer">
    <footer align="center">
      <p>Footer : <a href="about.html">About</a> | <a href="contact.html">Contact</a> |
        Copyright &copy; StudentHub Portal</p>
    </footer>
  </div>
  <script src="script.js"></script>
</body>

</html>
```

HTML Code - login.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
```

```
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="register.html">Register</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Login Box -->
    <div class="auth-card">
      <h2>LOGIN</h2>

      <form action="dashboard.html">
        <div class="form-group">
          <label for="email">Email Address</label>
          <input type="email" id="email" required placeholder="Enter your email">
        </div>

        <div class="form-group">
          <label for="password">Password</label>
          <input type="password" id="password" required placeholder="Enter your password">
        </div>

        <div class="form-options">
          <label for="remember">
            <input type="checkbox" id="remember"> Remember Me
          </label>
          <a href="#">Forgot Password?</a>
        </div>

        <button type="submit" class="auth-btn">Login</button>

        <p class="auth-footer-text">
          Don't have an account? <a href="register.html">Register here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - register.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="login.html">Login</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Student Registration Box -->
    <div class="auth-card register-card">
      <h2>STUDENT REGISTRATION</h2>

      <form id="registrationForm" onsubmit="return validateRegistration(event)">
        <div class="form-grid">
          <!-- 1. Full Name -->
          <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" placeholder="e.g. John Doe">
            <small class="error-text" id="nameError"></small>
          </div>

          <!-- 2. Email -->
          <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" placeholder="student@college.edu">
            <small class="error-text" id="emailError"></small>
          </div>

          <!-- 3. Mobile Number -->
          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" placeholder="10-digit mobile number">
            <small class="error-text" id="mobileError"></small>
          </div>

          <!-- 4. Course -->
          <div class="form-group">
            <label for="course">Course</label>
            <select id="course">
              <option value="">-- Select Course --</option>
              <option value="B.Tech Computer Science">B.Tech Computer Science</option>
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="BCA">BCA</option>
              <option value="MCA">MCA</option>
              <option value="B.Sc IT">B.Sc IT</option>
            </select>
            <small class="error-text" id="courseError"></small>
```

```
          </div>

          <!-- 5. Year -->
          <div class="form-group">
            <label for="year">Year of Study</label>
            <input type="number" id="year" min="1" max="4" placeholder="Enter Year (1 - 4)">
            <small class="error-text" id="yearError"></small>
          </div>

          <!-- 6. Gender -->
          <div class="form-group">
            <label>Gender</label>
            <div class="radio-group">
              <label for="male">
                <input type="radio" name="gender" id="male" value="male"> Male
              </label>
              <label for="female">
                <input type="radio" name="gender" id="female" value="female"> Female
              </label>
              <label for="other">
                <input type="radio" name="gender" id="other" value="other"> Other
              </label>
            </div>
            <small class="error-text" id="genderError"></small>
          </div>

          <!-- 7. Password -->
          <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Minimum 6 chars (letters &
numbers)">
            <small class="error-text" id="passwordError"></small>
          </div>

          <!-- 8. Confirm Password -->
          <div class="form-group">
            <label for="confirmPassword">Confirm Password</label>
            <input type="password" id="confirmPassword" placeholder="Re-type password">
            <small class="error-text" id="confirmPasswordError"></small>
          </div>
        </div>

        <!-- 9. Terms Acceptance -->
        <div class="form-options center-terms">
          <label for="terms">
            <input type="checkbox" id="terms"> I accept the Terms &amp;amp; Conditions
          </label>
          <small class="error-text" id="termsError"></small>
        </div>

        <button type="submit" class="auth-btn">Register</button>

        <p class="auth-footer-text">
          Already registered? <a href="login.html">Login here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
```

```
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - dashboard.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="style.css">
  <title>Student Hub Portal - Dashboard</title>
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Dashboard</strong>
    </div>

    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1000px; margin: 0 auto;">
    <div style="display: flex; gap: 20px;">
      <!-- Sidebar Navigation -->
      <div class="table1" style="flex: 1; max-width: 250px;">
        <div class="tablebutton">
          <p><a href="index.html"><button>Home</button></a></p>
          <p><a href="profile.html"><button>Profile</button></a></p>
          <p><a href="events.html"><button>Events</button></a></p>
          <p><a href="login.html"><button>Logout</button></a></p>
        </div>
      </div>

      <!-- Main Dashboard Content -->
      <div style="flex: 3;">
        <h2 style="font-size: 28px; margin-bottom: 15px;">Welcome, Student</h2>

        <!-- Stat Cards -->
        <div style="display: flex; gap: 15px; margin-bottom: 15px;">
          <div class="table1" style="flex: 1; text-align: center;">
            <p style="font-size: 20px; color: #EC9F05;">Total Events</p>
            <h1 style="font-size: 36px;">20</h1>
```

```
          </div>
          <div class="table1" style="flex: 1; text-align: center;">
            <p style="font-size: 20px; color: #EC9F05;">Registered</p>
            <h1 style="font-size: 36px;">5</h1>
          </div>
        </div>

        <!-- Activity Cards -->
        <div style="display: flex; gap: 15px;">
          <div class="table1" style="flex: 1; text-align: center;">
            <h3 style="color: #EC9F05; margin-bottom: 8px;">Upcoming Event</h3>
            <p><strong>AI &amp; Robotics Workshop</strong></p>
            <p style="font-size: 14px; margin-top: 5px;">Nov 10, 2026</p>
          </div>
          <div class="table1" style="flex: 1; text-align: center;">
            <h3 style="color: #EC9F05; margin-bottom: 8px;">Latest Notice</h3>
            <p><strong>Semester Examination</strong></p>
            <p style="font-size: 14px; margin-top: 5px;">Schedule published</p>
          </div>
        </div>
      </div>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - profile.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Profile</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
```

```
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="profile.html">Profile</a>
    </nav>
  </div>

  <main align="center" style="max-width: 500px; margin: 20px auto; padding: 10px;">
    <div class="table1" style="padding: 30px;">
      <!-- Profile Picture Circle Box -->
      <div style="width: 110px; height: 110px; border-radius: 50%; background-color: #6B4EFD;
color: white; display: flex; align-items: center; justify-content: center; margin: 0 auto;
border: 3px solid #EC9F05; font-weight: bold; text-align: center;">
        Profile<br>Picture
      </div>

      <hr style="border-color: #6B4EFD; margin: 20px 0;">

      <table width="100%" cellpadding="8" style="text-align: left; font-size: 16px;">
        <tr>
          <td width="35%" style="color: #EC9F05;"><strong>Name :</strong></td>
          <td>John Doe</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Email :</strong></td>
          <td>john.doe@studenthub.edu</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Mobile :</strong></td>
          <td>+1 987 654 3210</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Course :</strong></td>
          <td>B.Tech Computer Science</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Semester :</strong></td>
          <td>5th Semester</td>
        </tr>
      </table>
      <br>
      <button style="padding: 10px 25px; font-size: 16px;">Edit Profile</button>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - admin.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Admin Panel</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" height="45"
          style="vertical-align: middle; margin-right: 8px;"></a>
      <strong style="margin: 10px;">Admin Panel</strong>
    </div>
    <div class="headerlinks">
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <nav align="center">
      <a href="admin.html">Dashboard</a> |
      <a href="events.html">Events</a> |
      <a href="index.html">Home</a>
    </nav>
  </div>

  <main style="max-width: 1000px; margin: 20px auto; padding: 10px;">

    <!-- Stats Grid -->
    <div style="display: flex; gap: 15px; flex-wrap: wrap;">
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px; padding:
20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Students</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">250</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px; padding:
20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Events</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">12</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px; padding:
20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Feedback</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">18</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px; padding:
20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Notices</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">7</h1>
      </div>
    </div>

    <br>

    <!-- Recent Activities Card -->
    <div class="table1" style="padding: 25px;">
      <h3 style="font-size: 26px; color: #EC9F05; margin-top: 0;">Recent Activities</h3>
      <ul style="font-size: 18px; line-height: 1.8;">
        <li>Student Registered</li>
        <li>Event Added</li>
        <li>Feedback Submitted</li>
      </ul>
    </div>
  </main>
  <br>
  <div class="footer">
    <footer align="center">
      <p>Footer : <a href="about.html">About</a> | <a href="contact.html">Contact</a> |
```

```
        Copyright &copy; StudentHub Portal</p>
    </footer>
  </div>
</body>

</html>
```

Practical No. 3 - Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Question: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Ensure mobile-first responsiveness.

Key Questions

Q: How does the layout adapt for mobile, tablet, and desktop screens?

Answer: Flexbox allows items to wrap and share available space, while media queries change widths and the navigation layout for smaller screens. The same content can fit different screen sizes.

Q: Which layout technique is used and why?

Answer: Flexbox is used for horizontal groups such as the header, navigation, form grid, and cards. It is suitable because it aligns items and distributes space with less code.

Q: Are typography, spacing, colors, and contrast readable?

Answer: The stylesheet uses a dark background, light text, contrasting yellow buttons, consistent padding, and readable font sizes. These choices make the important content visible.

Q: Is the framework used consistently without unnecessary inline CSS?

Answer: The project mainly uses one external style.css file. Some page-specific inline styles are present, but the repeated site-wide styles are kept in the common stylesheet.

Analysis

A: The CSS file defines the common color palette, cards, buttons, forms, navigation, and responsive rules. Flexbox is used for groups of elements and media queries adjust the layout for smaller screens.

Interpretation

I: The interface is designed to remain usable on desktop and mobile screens. Centralizing the main styles makes the pages look like one portal instead of separate unrelated pages.

CSS Code - style.css

```
/* ================================================================================
   STUDENT HUB PORTAL - RESPONSIVE MOBILE-FIRST CSS (GRID & FLEXBOX)
   ================================================================================ */

/* --------------------------------------------------------------------------------
   1. BASE & RESET STYLES
   -------------------------------------------------------------------------------- */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  background-color: #0B162C;
  color: #FFFFFF;
  font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
  line-height: 1.6;
  padding: 12px;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  transition: background-color 0.3s, color 0.3s;
```

```
}

main {
  flex: 1;
}

/* Light Theme Variables & Styles */
body.light-theme {
  background-color: #F0F2F5;
  color: #1A202C;
}

body.light-theme .header strong,
body.light-theme label,
body.light-theme .auth-card h2,
body.light-theme .auth-footer-text {
  color: #1A202C;
}

body.light-theme .box,
body.light-theme .boxes2,
body.light-theme .table1,
body.light-theme .footer,
body.light-theme .modal-box,
body.light-theme .slider-container,
body.light-theme .auth-card {
  background-color: #FFFFFF;
  color: #1A202C;
  border: 1px solid #E2E8F0;
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.08);
}

body.light-theme .t1links a {
  color: #1A202C;
}

body.light-theme input[type="text"],
body.light-theme input[type="email"],
body.light-theme input[type="password"],
body.light-theme input[type="number"],
body.light-theme input[type="tel"],
body.light-theme select,
body.light-theme textarea {
  background-color: #F8FAFC;
  color: #1A202C;
  border: 1.5px solid #CBD5E1;
}

body.light-theme input:focus,
body.light-theme select:focus,
body.light-theme textarea:focus {
  background-color: #FFFFFF;
  border-color: #6B4EFD;
}

a {
  color: #EC9F05;
  text-decoration: none;
  transition: 0.2s ease;
}

a:hover {
  color: #d88e04;
  text-decoration: underline;
}
```

```
/* ------------------------------------------------------------------------
   2. LOGO & HEADER STYLES (FIXED SIZE ON ALL WEBSITES)
   ------------------------------------------------------------------------ */
.header img,
.logo,
header img {
  height: 48px !important;
  max-height: 48px !important;
  width: auto !important;
  max-width: 160px !important;
  object-fit: contain;
  vertical-align: middle;
  border-radius: 8px;
}

header {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: space-between;
  gap: 15px;
  padding: 10px 0;
}

.header {
  display: flex;
  align-items: center;
  gap: 12px;
  font-family: serif;
  font-size: 24px;
}

.header strong {
  color: #FFFFFF;
}

.headerlinks {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  justify-content: center;
  align-items: center;
}

.headerlinks button,
.headerlinks a button {
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 20px;
  font-size: 15px;
  font-weight: 600;
  padding: 8px 18px;
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s;
}

.headerlinks button:hover {
  background-color: #d88e04;
  transform: translateY(-1px);
}

.headerlinks a {
  color: white;
  text-decoration: none;
}
```

```css
/* ------------------------------------------------------------------------
   3. NOTIFICATION BANNER
   ------------------------------------------------------------------------ */
.notification-banner {
  background-color: #6B4EFD;
  color: white;
  padding: 10px 18px;
  border-radius: 12px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 12px;
  font-size: 15px;
}

.banner-close {
  background: none;
  border: none;
  color: white;
  font-size: 20px;
  font-weight: bold;
  cursor: pointer;
  padding: 0 6px;
}

/* ------------------------------------------------------------------------
   4. NAVIGATION BAR & HAMBURGER (FLEXBOX)
   ------------------------------------------------------------------------ */
.hamburger {
  display: block;
  background-color: #6B4EFD;
  color: white;
  border: none;
  border-radius: 10px;
  font-size: 18px;
  font-weight: bold;
  padding: 8px 16px;
  cursor: pointer;
  margin: 0 auto;
}

.navlinks {
  background-color: #6B4EFD;
  border-radius: 15px;
  margin: 10px 0 20px 0;
  padding: 10px;
}

.navlinks nav {
  display: none;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  gap: 10px;
}

.navlinks nav.show {
  display: flex;
}

.navlinks a {
  font-family: sans-serif;
  color: #FFFFFF;
  font-size: 16px;
  font-weight: 500;
  text-decoration: none;
  padding: 6px 12px;
  border-radius: 8px;
```

```
  transition: all 0.2s ease;
}

.navlinks a:hover {
  color: #DAF264;
  background-color: rgba(255, 255, 255, 0.15);
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   5. PERFECT LOGIN & REGISTRATION BOXES + FORM VALIDATION
   ------------------------------------------------------------------------ */
.auth-card {
  background-color: #212C37;
  border: 1px solid rgba(255, 255, 255, 0.08);
  border-radius: 24px;
  padding: 35px 28px;
  max-width: 440px;
  width: 100%;
  margin: 25px auto;
  box-shadow: 0 12px 30px rgba(0, 0, 0, 0.35);
  box-sizing: border-box;
}

.auth-card.register-card {
  max-width: 680px;
}

.auth-card h2 {
  font-size: 28px;
  color: #EC9F05;
  text-align: center;
  margin-bottom: 22px;
  letter-spacing: 0.5px;
}

.form-group {
  display: flex;
  flex-direction: column;
  gap: 6px;
  margin-bottom: 16px;
  text-align: left;
}

.form-group label {
  font-size: 15px;
  font-weight: 600;
  color: #E2E8F0;
}

.form-group input,
.form-group select,
.form-group textarea {
  width: 100%;
  background-color: #0B162C;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  border-radius: 12px;
  padding: 11px 15px;
  font-size: 15px;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  box-sizing: border-box;
}

.form-group input:focus,
.form-group select:focus,
```

```
.form-group textarea:focus {
  border-color: #EC9F05;
  box-shadow: 0 0 0 3px rgba(236, 159, 5, 0.2);
}

/* Error & Success States */
.error-text {
  color: #FF5252;
  font-size: 13px;
  margin-top: 3px;
  display: block;
  min-height: 18px;
}

.input-error {
  border-color: #FF5252 !important;
  box-shadow: 0 0 0 2px rgba(255, 82, 82, 0.25) !important;
}

.input-success {
  border-color: #10B981 !important;
}

/* 2-Column Grid for Registration Form */
.form-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 0 20px;
}

.radio-group {
  display: flex;
  gap: 20px;
  align-items: center;
  padding: 10px 0 4px 0;
}

.radio-group label {
  display: flex;
  align-items: center;
  gap: 6px;
  cursor: pointer;
  font-weight: normal;
}

.radio-group input[type="radio"],
.form-options input[type="checkbox"] {
  width: 18px;
  height: 18px;
  accent-color: #EC9F05;
  cursor: pointer;
}

.form-options {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
  font-size: 14px;
}

.form-options.center-terms {
  flex-direction: column;
  align-items: center;
  justify-content: center;
  margin-top: 10px;
  text-align: center;
```

```
}

.form-options label {
  display: flex;
  align-items: center;
  gap: 8px;
  cursor: pointer;
}

.auth-btn {
  width: 100%;
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 30px;
  font-size: 17px;
  font-weight: bold;
  padding: 13px;
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s;
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.3);
}

.auth-btn:hover {
  background-color: #d88e04;
  transform: translateY(-2px);
}

.auth-footer-text {
  text-align: center;
  margin-top: 18px;
  font-size: 15px;
  color: #CBD5E1;
}

/* ------------------------------------------------------------------------
   6. HERO, CARDS & GRIDS (HOME & ABOUT)
   ------------------------------------------------------------------------ */
.main {
  background-color: #6B4EFD;
  border-radius: 20px;
  padding: 30px 20px;
  margin: 15px 0;
  text-align: center;
  color: white;
}

.main h1 {
  font-size: 28px;
  margin-bottom: 8px;
}

.main p {
  font-size: 16px;
}

div[style*="display: flex"] {
  display: flex;
  flex-direction: column;
  gap: 15px;
  width: 100%;
}

.box,
.boxes2,
.table1 {
  background-color: #212C37;
```

```
  border-radius: 20px;
  border: 1px solid rgba(255, 255, 255, 0.05);
  padding: 24px;
  width: 100%;
  box-sizing: border-box;
}

.box {
  min-height: 220px;
  text-align: center;
}

.boxes2 {
  min-height: 180px;
  text-align: center;
}

.box h3,
.boxes2 p:first-child {
  color: #EC9F05;
  margin-bottom: 12px;
}

.box ul {
  list-style-position: inside;
  text-align: left;
  display: inline-block;
}

.t1,
.t1links {
  font-size: 17px;
  margin-bottom: 8px;
}

.t1links a {
  color: #FFFFFF;
}

.t1links a:hover {
  color: #EC9F05;
}

/* ------------------------------------------------------------------------
   7. COLLAPSIBLE FAQ & MODAL & SLIDER
   ------------------------------------------------------------------------ */
.faq-item {
  border-bottom: 1px solid #6B4EFD;
  padding: 10px 0;
}

.faq-btn {
  width: 100%;
  background: none;
  color: #EC9F05;
  font-size: 18px;
  font-weight: bold;
  border: none;
  padding: 10px 0;
  cursor: pointer;
  display: flex;
  justify-content: space-between;
  align-items: center;
  text-align: left;
}
```

```
.faq-answer {
  display: none;
  padding: 8px 0 12px 0;
  font-size: 16px;
}

.modal {
  display: none;
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: rgba(0, 0, 0, 0.75);
  justify-content: center;
  align-items: center;
  z-index: 1000;
  padding: 15px;
}

.modal-box {
  background-color: #212C37;
  color: white;
  padding: 30px 24px;
  border-radius: 20px;
  max-width: 480px;
  width: 100%;
  text-align: center;
  position: relative;
  border: 1px solid #6B4EFD;
}

.modal-close {
  position: absolute;
  top: 10px;
  right: 15px;
  background: none;
  color: white;
  font-size: 24px;
  border: none;
  cursor: pointer;
}

.slider-container {
  background-color: #212C37;
  border-radius: 20px;
  padding: 25px 20px;
  margin: 20px 0;
  text-align: center;
}

.slide {
  display: none;
}

.slide h3 {
  color: #EC9F05;
  font-size: 22px;
  margin-bottom: 8px;
}

.slider-controls {
  margin-top: 15px;
  display: flex;
  justify-content: center;
  gap: 12px;
}
```

```
/* ------------------------------------------------------------------------
   8. FOOTER STYLES
   ------------------------------------------------------------------------ */
.footer {
  background-color: #212C37;
  border-radius: 25px;
  padding: 16px;
  margin-top: 35px;
  text-align: center;
}

.footer footer p {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  justify-content: center;
  gap: 8px;
  font-size: 15px;
}

.footer a {
  background-color: #EC9F05;
  color: white;
  border-radius: 15px;
  font-size: 14px;
  padding: 4px 12px;
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   9. DESKTOP & TABLET RESPONSIVENESS
   ------------------------------------------------------------------------ */
@media (min-width: 768px) {
  body {
    padding: 20px 40px;
  }

  header {
    flex-direction: row;
    padding: 10px 10px;
  }

  .header {
    font-size: 30px;
  }

  .hamburger {
    display: none;
  }

  .navlinks nav {
    display: flex !important;
    flex-direction: row;
  }

  .main {
    max-width: 850px;
    margin: 20px auto;
    padding: 35px 20px;
  }

  .main h1 {
    font-size: 42px;
  }
```

```
  /* 2-Column Grid on Registration Form */
  .form-grid {
    grid-template-columns: 1fr 1fr;
  }

  /* Horizontal Grid on Desktop */
  div[style*="display: flex"] {
    flex-direction: row !important;
  }

  .box {
    width: 33.33% !important;
  }

  .boxes2 {
    width: 50% !important;
  }
}
```

Practical No. 4 - JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Question: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Key Questions

Q: How are DOM elements selected and modified?

Answer: The script uses getElementById, querySelector, and querySelectorAll. It changes class names, text, display styles, and the current slide to update the page.

Q: Are event listeners attached correctly?

Answer: The HTML buttons call functions for clicks, and a DOMContentLoaded listener starts the first slider. A window click handler also closes a modal when the user clicks outside it.

Q: Is localStorage used for remembering theme choice?

Answer: Yes. The selected theme is stored in localStorage, and the saved value is checked when the page loads so the preference can be restored.

Q: Does interactivity improve usability without breaking accessibility?

Answer: The features make the page easier to use by hiding repeated information, showing announcements, and supporting mobile navigation. Clear buttons and visible text keep the controls understandable.

Analysis

A: The JavaScript file contains small functions for theme switching, menu display, banner closing, FAQ expansion, modal display, and slider movement. Each function changes only the DOM elements needed for that feature.

Interpretation

I: The portal becomes interactive without a server. Users can control the menu, announcements, FAQ answers, theme, and slider directly in the browser.

JavaScript Code - script.js

```
/* ========================================================================
   STUDENT HUB PORTAL - SIMPLE DYNAMIC UI & FORM VALIDATION JAVASCRIPT
   ======================================================================== */


// 1. LIGHT / DARK THEME SWITCHER
function toggleTheme() {
  document.body.classList.toggle('light-theme');
  const isLight = document.body.classList.contains('light-theme');
  localStorage.setItem('theme', isLight ? 'light' : 'dark');
}


// Load saved theme on startup
if (localStorage.getItem('theme') === 'light') {
  document.body.classList.add('light-theme');
}


// 2. HAMBURGER MENU (MOBILE NAVIGATION)
function toggleMenu() {
  const menu = document.getElementById('nav-menu');
```

```
  if (menu) {
    menu.classList.toggle('show');
  }
}

// 3. NOTIFICATION BANNER
function closeBanner() {
  const banner = document.getElementById('notification-banner');
  if (banner) {
    banner.style.display = 'none';
  }
}

// 4. COLLAPSIBLE FAQ ACCORDION
function toggleFaq(btn) {
  const answer = btn.nextElementSibling;
  const icon = btn.querySelector('.faq-icon');

  if (answer.style.display === 'block') {
    answer.style.display = 'none';
    if (icon) icon.textContent = '+';
  } else {
    answer.style.display = 'block';
    if (icon) icon.textContent = '-';
  }
}

// 5. MODAL POPUP (OPEN / CLOSE)
function openModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'flex';
  }
}

function closeModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'none';
  }
}

// Close modal if user clicks outside the modal box
window.onclick = function(event) {
  if (event.target.classList.contains('modal')) {
    event.target.style.display = 'none';
  }
};

// 6. IMAGE / CONTENT SLIDER
let currentSlide = 0;

function showSlide(index) {
  const slides = document.querySelectorAll('.slide');
  if (slides.length === 0) return;

  if (index >= slides.length) {
    currentSlide = 0;
  } else if (index < 0) {
    currentSlide = slides.length - 1;
  } else {
    currentSlide = index;
  }

  slides.forEach((slide, i) => {
    slide.style.display = (i === currentSlide) ? 'block' : 'none';
```

```
  });
}

function changeSlide(step) {
  showSlide(currentSlide + step);
}

// Initialize first slide on page load
document.addEventListener('DOMContentLoaded', () => {
  showSlide(0);
});

// 7. STUDENT REGISTRATION FORM VALIDATION (REGEX)
function validateRegistration(event) {
  event.preventDefault(); // Stop normal form submission

  let isValid = true;

  // Helper functions for displaying error / success
  function setError(id, message) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = message;
    if (inputElem) {
      inputElem.classList.add('input-error');
      inputElem.classList.remove('input-success');
    }
    isValid = false;
  }

  function setSuccess(id) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = '';
    if (inputElem) {
      inputElem.classList.remove('input-error');
      inputElem.classList.add('input-success');
    }
  }

  // 1. Name Validation (Letters & spaces, 3-30 characters)
  const name = document.getElementById('name').value.trim();
  const nameRegex = /^[A-Za-z\s]{3,30}$/;
  if (!nameRegex.test(name)) {
    setError('name', 'Name must be 3-30 letters without numbers/symbols');
  } else {
    setSuccess('name');
  }

  // 2. Email Validation (Standard email pattern)
  const email = document.getElementById('email').value.trim();
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  if (!emailRegex.test(email)) {
    setError('email', 'Please enter a valid email address');
  } else {
    setSuccess('email');
  }

  // 3. Mobile Number Validation (10-digit mobile number)
  const mobile = document.getElementById('mobile').value.trim();
  const mobileRegex = /^[6-9]\d{9}$/;
  if (!mobileRegex.test(mobile)) {
    setError('mobile', 'Enter a valid 10-digit mobile number');
  } else {
    setSuccess('mobile');
  }
```

```
  // 4. Course Validation (Selection required)
  const course = document.getElementById('course').value;
  if (!course) {
    setError('course', 'Please select your course');
  } else {
    setSuccess('course');
  }

  // 5. Year Validation (1 to 4)
  const year = document.getElementById('year').value.trim();
  const yearRegex = /^[1-4]$/;
  if (!yearRegex.test(year)) {
    setError('year', 'Enter a valid year (1 to 4)');
  } else {
    setSuccess('year');
  }

  // 6. Gender Validation (Radio button checked)
  const genderChecked = document.querySelector('input[name="gender"]:checked');
  const genderError = document.getElementById('genderError');
  if (!genderChecked) {
    if (genderError) genderError.textContent = 'Please select your gender';
    isValid = false;
  } else {
    if (genderError) genderError.textContent = '';
  }

  // 7. Password Validation (At least 6 characters with letters and numbers)
  const password = document.getElementById('password').value;
  const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d@$!%*#?&]{6,}$/;
  if (!passwordRegex.test(password)) {
    setError('password', 'Password must have at least 6 characters (letters & numbers)');
  } else {
    setSuccess('password');
  }

  // 8. Confirm Password Validation (Must match password)
  const confirmPassword = document.getElementById('confirmPassword').value;
  if (confirmPassword === '' || confirmPassword !== password) {
    setError('confirmPassword', 'Passwords do not match');
  } else {
    setSuccess('confirmPassword');
  }

  // 9. Terms Acceptance (Checkbox must be checked)
  const terms = document.getElementById('terms');
  const termsError = document.getElementById('termsError');
  if (!terms.checked) {
    if (termsError) termsError.textContent = 'You must accept the Terms & Conditions';
    isValid = false;
  } else {
    if (termsError) termsError.textContent = '';
  }

  // Final Action on Successful Validation
  if (isValid) {
    alert('🎉 Student Registration Successful! Redirecting to Dashboard...');
    window.location.href = 'dashboard.html';
  }

  return isValid;
}
```

Practical No. 5 - Registration Form with Frontend Validation and User-Friendly Error Handling

Question: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Key Questions

Q: Are correct input types and attributes used?

Answer: Yes. The form uses text, email, tel, number, password, select, radio, and checkbox controls. Labels, placeholders, min and max values, and validation checks support correct input.

Q: Are validation errors displayed near the relevant fields?

Answer: Yes. Each important field has a nearby error element such as nameError or emailError. The validation function writes the message there and adds an input-error class.

Q: Is password strength checked?

Answer: Yes. A regular expression checks that the password has at least six characters, at least one letter, and at least one number. The confirm password is also compared with it.

Q: Is the form accessible using keyboard and screen-reader-friendly labels?

Answer: The controls are connected to visible labels using for and id values, and the radio buttons have a shared name. This gives keyboard and assistive-technology users understandable fields.

Analysis

A: The registration form collects the required student details and validates them before accepting the form. Invalid fields receive a message and an error class, while valid fields receive a success class.

Interpretation

I: Client-side validation gives immediate feedback and prevents incomplete or incorrectly formatted data from being submitted. Server-side validation would still be needed in a real application.

HTML Code - register.html

```
<!DOCTYPE html>
<html lang="en">


<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>


<body>


  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo" class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Login</a></button>
```

```
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="login.html">Login</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Student Registration Box -->
    <div class="auth-card register-card">
      <h2>STUDENT REGISTRATION</h2>

      <form id="registrationForm" onsubmit="return validateRegistration(event)">
        <div class="form-grid">
          <!-- 1. Full Name -->
          <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" placeholder="e.g. John Doe">
            <small class="error-text" id="nameError"></small>
          </div>

          <!-- 2. Email -->
          <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" placeholder="student@college.edu">
            <small class="error-text" id="emailError"></small>
          </div>

          <!-- 3. Mobile Number -->
          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" placeholder="10-digit mobile number">
            <small class="error-text" id="mobileError"></small>
          </div>

          <!-- 4. Course -->
          <div class="form-group">
            <label for="course">Course</label>
            <select id="course">
              <option value="">-- Select Course --</option>
              <option value="B.Tech Computer Science">B.Tech Computer Science</option>
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="BCA">BCA</option>
              <option value="MCA">MCA</option>
              <option value="B.Sc IT">B.Sc IT</option>
            </select>
            <small class="error-text" id="courseError"></small>
          </div>

          <!-- 5. Year -->
          <div class="form-group">
            <label for="year">Year of Study</label>
            <input type="number" id="year" min="1" max="4" placeholder="Enter Year (1 - 4)">
            <small class="error-text" id="yearError"></small>
          </div>

          <!-- 6. Gender -->
          <div class="form-group">
```

```
            <label>Gender</label>
            <div class="radio-group">
              <label for="male">
                <input type="radio" name="gender" id="male" value="male"> Male
              </label>
              <label for="female">
                <input type="radio" name="gender" id="female" value="female"> Female
              </label>
              <label for="other">
                <input type="radio" name="gender" id="other" value="other"> Other
              </label>
            </div>
            <small class="error-text" id="genderError"></small>
          </div>

          <!-- 7. Password -->
          <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Minimum 6 chars (letters &
numbers)">
            <small class="error-text" id="passwordError"></small>
          </div>

          <!-- 8. Confirm Password -->
          <div class="form-group">
            <label for="confirmPassword">Confirm Password</label>
            <input type="password" id="confirmPassword" placeholder="Re-type password">
            <small class="error-text" id="confirmPasswordError"></small>
          </div>
        </div>

        <!-- 9. Terms Acceptance -->
        <div class="form-options center-terms">
          <label for="terms">
            <input type="checkbox" id="terms"> I accept the Terms &amp;amp; Conditions
          </label>
          <small class="error-text" id="termsError"></small>
        </div>

        <button type="submit" class="auth-btn">Register</button>

        <p class="auth-footer-text">
          Already registered? <a href="login.html">Login here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

JavaScript Code - validation in script.js

```
/* ========================================================================
   STUDENT HUB PORTAL - SIMPLE DYNAMIC UI & FORM VALIDATION JAVASCRIPT
   ======================================================================== */
```

```
// 1. LIGHT / DARK THEME SWITCHER
function toggleTheme() {
  document.body.classList.toggle('light-theme');
  const isLight = document.body.classList.contains('light-theme');
  localStorage.setItem('theme', isLight ? 'light' : 'dark');
}

// Load saved theme on startup
if (localStorage.getItem('theme') === 'light') {
  document.body.classList.add('light-theme');
}

// 2. HAMBURGER MENU (MOBILE NAVIGATION)
function toggleMenu() {
  const menu = document.getElementById('nav-menu');
  if (menu) {
    menu.classList.toggle('show');
  }
}

// 3. NOTIFICATION BANNER
function closeBanner() {
  const banner = document.getElementById('notification-banner');
  if (banner) {
    banner.style.display = 'none';
  }
}

// 4. COLLAPSIBLE FAQ ACCORDION
function toggleFaq(btn) {
  const answer = btn.nextElementSibling;
  const icon = btn.querySelector('.faq-icon');

  if (answer.style.display === 'block') {
    answer.style.display = 'none';
    if (icon) icon.textContent = '+';
  } else {
    answer.style.display = 'block';
    if (icon) icon.textContent = '-';
  }
}

// 5. MODAL POPUP (OPEN / CLOSE)
function openModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'flex';
  }
}

function closeModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'none';
  }
}

// Close modal if user clicks outside the modal box
window.onclick = function(event) {
  if (event.target.classList.contains('modal')) {
    event.target.style.display = 'none';
  }
};

// 6. IMAGE / CONTENT SLIDER
```

```
let currentSlide = 0;

function showSlide(index) {
  const slides = document.querySelectorAll('.slide');
  if (slides.length === 0) return;

  if (index >= slides.length) {
    currentSlide = 0;
  } else if (index < 0) {
    currentSlide = slides.length - 1;
  } else {
    currentSlide = index;
  }

  slides.forEach((slide, i) => {
    slide.style.display = (i === currentSlide) ? 'block' : 'none';
  });
}

function changeSlide(step) {
  showSlide(currentSlide + step);
}

// Initialize first slide on page load
document.addEventListener('DOMContentLoaded', () => {
  showSlide(0);
});

// 7. STUDENT REGISTRATION FORM VALIDATION (REGEX)
function validateRegistration(event) {
  event.preventDefault(); // Stop normal form submission

  let isValid = true;

  // Helper functions for displaying error / success
  function setError(id, message) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = message;
    if (inputElem) {
      inputElem.classList.add('input-error');
      inputElem.classList.remove('input-success');
    }
    isValid = false;
  }

  function setSuccess(id) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = '';
    if (inputElem) {
      inputElem.classList.remove('input-error');
      inputElem.classList.add('input-success');
    }
  }

  // 1. Name Validation (Letters & spaces, 3-30 characters)
  const name = document.getElementById('name').value.trim();
  const nameRegex = /^[A-Za-z\s]{3,30}$/;
  if (!nameRegex.test(name)) {
    setError('name', 'Name must be 3-30 letters without numbers/symbols');
  } else {
    setSuccess('name');
  }

  // 2. Email Validation (Standard email pattern)
```

```
const email = document.getElementById('email').value.trim();
const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
if (!emailRegex.test(email)) {
  setError('email', 'Please enter a valid email address');
} else {
  setSuccess('email');
}

// 3. Mobile Number Validation (10-digit mobile number)
const mobile = document.getElementById('mobile').value.trim();
const mobileRegex = /^[6-9]\d{9}$/;
if (!mobileRegex.test(mobile)) {
  setError('mobile', 'Enter a valid 10-digit mobile number');
} else {
  setSuccess('mobile');
}

// 4. Course Validation (Selection required)
const course = document.getElementById('course').value;
if (!course) {
  setError('course', 'Please select your course');
} else {
  setSuccess('course');
}

// 5. Year Validation (1 to 4)
const year = document.getElementById('year').value.trim();
const yearRegex = /^[1-4]$/;
if (!yearRegex.test(year)) {
  setError('year', 'Enter a valid year (1 to 4)');
} else {
  setSuccess('year');
}

// 6. Gender Validation (Radio button checked)
const genderChecked = document.querySelector('input[name="gender"]:checked');
const genderError = document.getElementById('genderError');
if (!genderChecked) {
  if (genderError) genderError.textContent = 'Please select your gender';
  isValid = false;
} else {
  if (genderError) genderError.textContent = '';
}

// 7. Password Validation (At least 6 characters with letters and numbers)
const password = document.getElementById('password').value;
const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d@$!%*#?&]{6,}$/;
if (!passwordRegex.test(password)) {
  setError('password', 'Password must have at least 6 characters (letters & numbers)');
} else {
  setSuccess('password');
}

// 8. Confirm Password Validation (Must match password)
const confirmPassword = document.getElementById('confirmPassword').value;
if (confirmPassword === '' || confirmPassword !== password) {
  setError('confirmPassword', 'Passwords do not match');
} else {
  setSuccess('confirmPassword');
}

// 9. Terms Acceptance (Checkbox must be checked)
const terms = document.getElementById('terms');
const termsError = document.getElementById('termsError');
if (!terms.checked) {
  if (termsError) termsError.textContent = 'You must accept the Terms & Conditions';
  isValid = false;
} else {
```

```
    if (termsError) termsError.textContent = '';
  }

  // Final Action on Successful Validation
  if (isValid) {
    alert('🎉 Student Registration Successful! Redirecting to Dashboard...');
    window.location.href = 'dashboard.html';
  }

  return isValid;
}
```